

Web Design Level 3

1. How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?
 - A) By ignoring accessibility guidelines
 - B) By following web accessibility standards, using ARIA roles, and testing with screen readers and other assistive technologies**
 - C) By using only plain HTML
 - D) By minimizing the use of JavaScript
2. Which image format supports transparency and is commonly used for web graphics and images with transparent backgrounds?
 - A) PSD
 - B) PNG**
 - C) JPG
 - D) GIF
3. What is the function of the HTML style attribute?
 - A. It is used to add style to an HTML element**
 - B. It is used to uniquely identify some specific styles of some element
 - C. Both a & b
 - D. None of the above
4. What tag is used to render an image on a webpage?
 - A. img**
 - B. src
 - C. image
 - D. none of them
5. The most basic part of any HTML page is?
 - A. ASCII Text**
 - B. Binary Text
 - C. Text
 - D. None of the above
6. How many characters can be written in 1KB?
 - A. 1048
 - B. 1024**
 - C. 2048
 - D. None of them
7. Explain the difference between the ---- and ---- operators in JavaScript.
 - A) ---- compares values; ---- compares values and types**
 - B) ---- compares values and types; ---- compares only values
 - C) ---- compares objects;---- compares arrays
 - D) ---- compares strings;---- compares numbers
8. What are logical operators?
 - A) +, -, , /
 - B) &&, ||, !**
 - C) =, !=, ==, !==
 - D) if, else switch
9. How do you use the break statement in loops and switches in JavaScript?

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- A) To pause execution
 - B) To terminate a loop or switch**
 - C) To skip an iteration
 - D) To execution
10. How do you display output in JavaScript?
- A) Using print()
 - B) Using alert()
 - C) Using console.log()**
 - D) Using document.write()
11. What are single-line and multi-line comments in JavaScript? How do you comment out code?
- A) // for single-line, /* for multi-line**
 - B) <!--> for single-line, <!--> for multi-line
 - C) # for single line, “””” “””” for Multi-line
 - D) // for single line, // for multi line
12. Select the basic data types in JavaScript.
- A) Number, String , Boolean, Object, Array
 - B) Number, String , Boolean, Undefined, Null, Object**
 - C) Integer, Float, String, Boolean, Object
 - D) Char, Float, String, Boolean, Array
13. How do you check the data type of a variable in JavaScript?
- A) Type of variable**
 - B) instance of variable
 - C) variable. type
 - D) check Type(variable)
14. What are arithmetic operators in JavaScript? Provide examples
- A) +,-,*,/**
 - B) +, -, , /, && ,
 - C) ++, --, +=, -=
 - D) ==, ===, !=, !==
15. Which keyword is used to declare a variable in JavaScript?
- A) var**
 - B) variable
 - C) v
 - D) let
16. Which operator is used for strict equality comparison in JavaScript?
- A) ==
 - B) !=
 - C) ===**
 - D) !==
17. Which function is used to output data to the console in JavaScript?
- A) print()
 - B) log()

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- C) display()
- D) console.log()**

18. Which built-in method is used to convert a string to an integer in JavaScript?
- A) parseInt()**
 - B) tointeger()
 - C) convert ToInt()
 - D) stringToInt()
19. What is the purpose of Tailwind CSS and how does it differ from other CSS frameworks like Bootstrap?
- A) Tailwind CSS is a JavaScript framework
 - B) Tailwind CSS is a utility-first CSS framework that provides low-level utility-----, while Bootstrap provides pre-designed components**
 - C) Tailwind CSS is used for server-side scripting
 - D) Tailwind CSS is a database management tool
20. What is the basic difference between JavaScript and Java?
- A. Functions are considered as fields
 - B. Functions are values, and there is no hard distinction between methods and fields**
 - C. Variables are specific
 - D. There is no difference
21. Which statement is used to execute a block of code multiple times in JavaScript?
- A. for**
 - B. if
 - C. return
 - D. break
22. Why event handlers are needed in JS?
- A. Allows JavaScript code to alter the behavior of windows**
 - B. Adds inner HTML page to the code
 - C. Change the server location
 - D. Performs handling of exceptions and occurrences
23. What is the purpose of JavaScript in web development?
- A. To structure a web page
 - B. To style web page
 - C. To add interactivity and dynamic content to web pages**
 - D. To store data on the server
24. In JavaScript, which of the following is a valid variable name?
- A. 2names
 - B. \$name**
 - C. -name
 - D. @name
25. Which of the following class is used to create a responsive navigation bar in Bootstrap?
- A. navbar-fixed
 - B. navbar-expand**

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- C. navbar-light
- D. navbar-collapse

26. What is the purpose of the card class in Bootstrap?

- A. To create interactive buttons.
- B. To apply borders to an element.
- C. To create a flexible content container with optional headers and footers.**
- D. To align text within a grid.

27. How can we write comments in CSS?

- A. //**
- B. //
- C. #
- D. All of this

28. The CSS property is used to control the transparency of an element?

- A. Opacity.**
- B. Visibility
- C. filter
- D. None of them

29. Which of the following CSS property is used for controlling the layout?

- A. display**
- B. color
- C. text
- D. None of them

30. Which of the following components of the CSS box model are transparent?

- A. padding
- B. margin
- C. both a & b**
- D. content

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31. How many types of layout are available in Bootstrap?

There are two major layouts for Bootstrap:

- A) Fluid Layout- This layout is necessary for creating an app that is 100 % wider and covers all the screen width.
- B) Fixed Layout- It is used only for a standard screen (940px). Both layouts can be used for creating a responsive design.

32. Write down the boilerplate of HTML?

```
<!DOCTYPE html>
<html>
  <head>
    <meta charact=" UTF-8"/>
    <title>Document Title</title>
  </head>
  <body>
  </body>
</html>
```

33. Why we use CSS box-sizing properly?

Box sizing property remove Border & Padding from height & width property in CSS

34. What is the difference between margin & padding?

- Margin is used to create space around elements and padding is used to create space around elements inside the border.
- We can set the margin property to auto but we cannot set the padding property to auto.
- In Margin property we can allow negative or float number but in padding we cannot allow negative values.

35. What is JSON & give an example JSON?

- JSON is a standard data interchange format that's used to store simple data structures and objects. It's a lightweight, language-independent format that's easy to learn and troubleshoot
- Var student = {name: "Rubel", city: "Dhaka" }

36. What are the timing method in JavaScript & Why we use it?

There are two timing method in JavaScript. Setinterval & Settimeout. These method is used for creating task repeatedly. Setinterval execute a task repeatedly within given time period.

Settimeout execute a task one time within given time period.

37. What is the difference between DOM & BOM?

- DOM stands for Document Object Model. DOM is a programming interface for HTML and XML documents.
- Browser Object Model is known as BOM. It allows users to interact with the browser. A browser's initial object is a window. As a result, you may call all of the window's functions directly or by referencing the window.

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38. What are the difference between HTML TAG, ELEMENT & ATTRIBUTE?

- HTML Tag is a syntax that is used to print data & define data print.
- HTML Element contain starting tag & ending tag with content.
- HTML Attribute provide additional information about HTML Tag

39. Describe different type HTML Element?

- There are two type of HTML Element,
- Block Label Element & Inline Element.
- Block Label element hold the whole area.
- Inline element hold only contain area.

40. What is function in JavaScript? Give an example?

Function is a block of code that has a particular job. Function can not work itself. When we call a function then it will work. Function can reuse.

```
function hello()
{
    Document.write("Hello World.");
}
```

41. What are Array in JavaScript? Give an example?

- Array is a special variable in JavaScript. Array can store multiple data. In JavaScript it is possible to store different type of data in an Array.
- Var data = [10, 20, "Hello World.", 20.34, true, false];

42. Create a JSON Array for that contain four students name, email, mobile & address.

```
Var students = [
    {name: "Arif", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
    {name: " Arif ", email: arif@gmail.com, mobile: "01313363836", address: "Dhaka"},
]
```

43. Describe different type of repeated statement in JavaScript?

- In JavaScript there are for, while, do while, foreach, forin repeated statement.
- For & while syntax start works when condition is true. Do while syntax one time works after check condition, if the condition is true then it works again.
- Foreach & Forin syntax works for Array.

44. Describe different type of conditional statements syntax in JavaScript?

- There are four conditional statement syntax in JavaScript. They are if, if else, if else if & switch.
- If syntax work if the condition is true.
- If else syntax work if the condition is true & if not true then it show another result.

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- If else if syntax work for multiple condition.
- Switch syntax work for property value. Inside switch syntax there are lot of case define. Which property value matches with case then that particular case will start execute statement & it stop after break syntax.

45. Create a registration form using only HTML. Form contain user name, email, mobile & address?

```
<form>
  <div>
    <label>User Name</label>
    <input type="text" name="name"/>
  </div>
  <div>
    <label>Email</label>
    <input type="email" name="email"/>
  </div>
  <div>
    <label>Mobile</label>
    <input type="number" name="mobile"/>
  </div>
  <div>
    <label>Address</label>
    <textarea name="address"></textarea>
  </div>
</form>
```

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46. What is the difference between inner Text and inner HTML in JavaScript?

- **innerText**: Gets or sets the text content of an element, excluding hidden elements. It respects CSS styling (like `display: none` and `visibility: hidden`).
- **innerHTML**: Gets or sets the HTML content inside an element, including tags. It parses the string as HTML, allowing for rendering of nested elements.

47. How does an interactive website differ from static and namic website?

- **Static Website**: Contains fixed content (HTML/CSS). It doesn't change without manual updates and has no user interaction (e.g., simple portfolio sites).
- **Dynamic Website**: Displays dynamic content using server-side scripting (e.g., PHP, Node.js). It fetches data from databases, allowing content updates without modifying the code (e.g., blogs, news sites).
- **Interactive Website**: Goes beyond dynamic content by engaging users through real-time interactions, animations, and user-driven experiences. It uses client-side scripting (JavaScript) for features like forms, chatbots, and dynamic navigation (e.g., social media platforms, web apps).

48. Which code editor do you prefer for web design and development projects, and why?

I prefer **Visual Studio Code (VS Code)** because it's fast, highly customizable, and has great extensions for HTML, CSS, and JavaScript. It also offers built-in Git integration, smart autocompletion, and a Live Server for real-time previews, making web development efficient and seamless.

49. Can you explain the basic purpose of FTP (File Transfer Protocol) software in web development?

FTP (File Transfer Protocol) software is used in web development to **upload, download, and manage files** on a web server. It allows developers to transfer website files (like HTML, CSS, images) from their local computer to the server, making the site accessible online. It's also used for updating and maintaining websites.

50. How do you create HTML login form with submit button?

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Form</title>
</head>
<body>
  <h2>Login Form</h2>
  <form action="login.php" method="post">
    <label for="username">Username:</label>
    <input type="text" id="username" name="username" required><br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" required><br><br>

    <button type="submit">Login</button>
  </form>
</body>
```

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</html>

51. When should I consider using SVG image?

Use ****SVG**** when you need scalable, high-quality graphics that are lightweight, editable with CSS/JavaScript, and accessible for SEO. Ideal for logos, icons, and simple illustrations.

52. What is an ID selector in CSS?

An ID selector in CSS is used to style a specific element with a unique identifier. It's defined using a # followed by the element's ID. Example:

```
#header {  
    color: blue;  
    font-size: 20px;  
}
```

53. How do you apply multiple CSS classes to a single HTML element?

```
<div class="class1 class2 class3">  
    Content here  
</div>
```

54. How do you create functions in JavaScript?

Function Declaration:

```
function greet(name) {  
    return "Hello, " + name;  
}
```

Arrow Function:

```
const greet = (name) => {  
    return "Hello, " + name;  
}
```

55. How do you create an ordered list in HTML?

To create an ordered list in HTML, use the (ordered list) tag, with each list item wrapped in an (list item) tag:

```
<ol>  
    <li>First item</li>  
    <li>Second item</li>  
    <li>Third item</li>  
</ol>
```

56. What are arrays and objects in JavaScript?

- **Arrays:** Arrays in JavaScript are used to store multiple values in a single variable. They are ordered collections and can hold any type of data.
- **Objects:** Objects are collections of key-value pairs, where each key is a string (or symbol) and the value can be any data type.

57. How do you access arrays and objects and their elements properties?

- ****Array**:** Access elements using an index, e.g., `fruits[0]`.
- ****Object**:** Access properties using dot or bracket notation, e.g., `person.name` or `person["age"]`.

58. Can you explain the difference between for loop, while loop, and do-while loop?

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- ****for loop****: Used when the number of iterations is known in advance.
- ****while loop****: Runs as long as the condition is true, checked before each iteration.
- ****do-while loop****: Runs at least once, condition checked after each iteration.

59. How do you handle conditional statements in JavaScript?

Use if to check a condition, else for the alternative, and else if for multiple conditions. The switch statement checks different values.

```
if (x > 10) { console.log("x > 10"); }  
else if (x === 10) { console.log("x = 10"); }  
else { console.log("x < 10"); }
```

60. What is the Document Object Model (DOM) in JavaScript?

The Document Object Model (DOM) is a programming interface for web documents. It represents the page as a tree structure, where each element, attribute, and text is a node. JavaScript can manipulate the DOM to change the content, structure, and style of a webpage dynamically.

For example, you can use JavaScript to change the text inside a <div>:

```
document.getElementById("myDiv").innerHTML = "New content!";
```

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